Connecting to external services

When you work with streams, you almost always need to call an external service: an HTTP service, a gRPC service, a database call, an FTP server, and many more. Let's look at some of these services.

In most streams, there is a moment when you need to enrich or validate the data through an HTTP call. To test this, you can use an Akka HTTP server where you check whether the quantity you pass it is greater than a number. You can find such a server in `HttpServer.scala`. The endpoint you have available is `/validate?quantity =[x]`, which responds with a validation class `Validated(accepted: Boolean)`. To start this server, enter this:

```
sbt "chapter16/runMain example.validation.HttpServer"
```

Once the server is running, you need the following:

- `RootJsonFormat`—So it can unmarshal the incoming result from the server.
- `Unmarshal`—To wrap the request and turn it into `Validated`.
- `Throttle`—The HTTP server has a maximum number of allowed connections. The default setting is HTTP `max-open-requests = 32`, from the reference .conf file of the `akka-http-core` library.

To call the external HTTP service, use an `Http().singleRequest` and an `HttpRequest`. To build the